

Qns	Answer	Marks
6a	The <u>increase in internal energy</u> of a system is equal to the <u>sum</u> of the <u>heat supplied to</u> the system and the <u>work done on</u> the <u>system</u> .	B2
6b	Since the temperature increases by the same amount for the same amount of gas, <u>the increase in internal energy (ΔU) is the same</u>. (First process) When heated at <u>constant volume</u>, <u>no work is done</u> on the gas ($W=0$). By first law of thermodynamics, heat transferred to the system is solely to increase its internal energy ($Q=\Delta U$). (Second process) When heated at constant pressure, the gas expands, and work is done by the gas ($W<0$). By first law of thermodynamics, <u>heat is required for the increase of internal energy AND the work done by the gas</u> ($\Delta U=Q+W \Rightarrow Q = \Delta U - W > \Delta U$, since $W<0$). Hence, to increase the temperature by the same amount, the heat required at constant pressure is greater than heat required at constant volume.	B1 B1 B1
6ci	Using equation of state for ideal gas: $pV = NkT$ $(2.62 \times 10^5)(0.0360) = N(1.38 \times 10^{-23})(273.15 + 25)$ $N = 2.29 \times 10^{24} = 2.3 \times 10^{24}$	M1 A1
6cii	mass per molecule $m = \frac{6.5 \times 10^{-3}}{6.02 \times 10^{23}}$ By kinetic theory, $pV = \frac{1}{3}Nm \langle c^2 \rangle$ $(2.62 \times 10^5)(0.0360) = \frac{1}{3} \times 2.29 \times 10^{24} \times \frac{6.5 \times 10^{-3}}{6.02 \times 10^{23}} \langle c^2 \rangle$ $c_{rms} = 1070 \text{ m s}^{-1}$	M1 (or finding n) M1 A1

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7a	Electric potential at a point is the work done (by an external agent) per	B1